



2012 Trial Examination

FORM VI

MATHEMATICS EXTENSION 1

Thursday 9th August 2012

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 70 Marks

- All questions may be attempted.

Section I – 10 Marks

- Questions 1–10 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown in every question.
- Start each question in a new booklet.

Collection

- Write your candidate number clearly on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eleven.
- Write your candidate number on this question paper and submit it with your answers.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 128 boys

Examiner
SO/MLS

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

What is the remainder when $P(x) = x^2 + 5x + 7$ is divided by $(x + 3)$?

1

(A) $-\frac{109}{9}$

(B) $\frac{19}{9}$

(C) 31

(D) 1

QUESTION TWO

The point R divides the interval joining $P(a, 2b)$ and $Q(3a, -b)$ externally in the ratio 2 : 3. What are the coordinates of R ?

1

(A) $(-3a, 8b)$

(B) $\left(\frac{11a}{5}, \frac{4b}{5}\right)$

(C) $(7a, -7b)$

(D) $\left(\frac{9a}{5}, \frac{8b}{5}\right)$

QUESTION THREE

The term independent of x in the expansion of $\left(x + \frac{2}{x}\right)^6$ is:

1

(A) 160

(B) 80

(C) 40

(D) 20

Exam continues next page ...

QUESTION FOUR

A simplified expression for $\binom{n+1}{n-1}$ is:

1

(A) $\frac{1}{2}(n^2 - n)$

(B) $\frac{1}{2}(n^2 + n)$

(C) $n^2 - n$

(D) $n^2 + n$

QUESTION FIVE

It is given that x_1 is a good approximate solution of $\cos x = x$. Using one step of Newton's method, a better approximation is:

1

(A) $x_2 = x_1 - \frac{\cos x_1 + x_1}{\sin x_1 + 1}$

(B) $x_2 = x_1 - \frac{\cos x_1 - x_1}{\sin x_1 - 1}$

(C) $x_2 = x_1 + \frac{\cos x_1 - x_1}{-\sin x_1 + 1}$

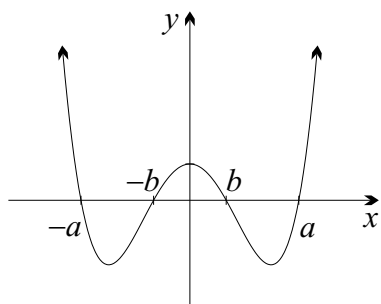
(D) $x_2 = x_1 + \frac{\cos x_1 - x_1}{\sin x_1 + 1}$

QUESTION SIX

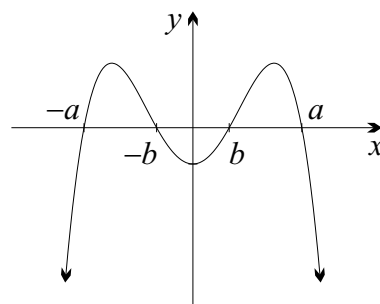
Which diagram best represents $P(x) = (x - a)^2(b^2 - x^2)$, where $a > b$?

1

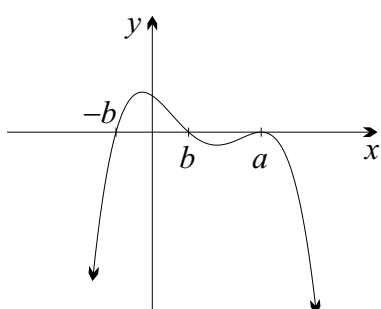
(A)



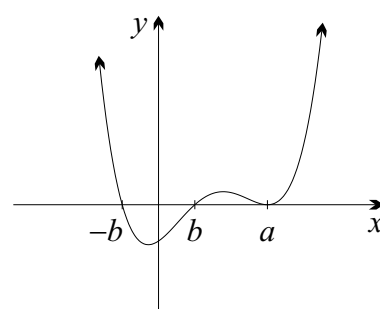
(B)



(C)



(D)



QUESTION SEVEN

What is the inverse function of $f(x) = \frac{5 + e^{2x}}{3}$?

1

(A) $\frac{3}{5 + e^{2x}}$

(B) e^{5-3x}

(C) $\frac{1}{2} \ln(3x - 5)$

(D) $\frac{1}{2} \ln(5x - 3)$

QUESTION EIGHT

Consider the polynomial $P(x) = x^4 + ax^3 + bx^2 + cx + d$, where a, b, c and d are real numbers. One of the zeros of this polynomial is 1. What is the value of $a + b + c + d$? 1

- (A) -1
- (B) $-1 - \alpha^2$
- (C) $-2 - \alpha^2$
- (D) 0

QUESTION NINE

An expression for the general solution to the trigonometric equation $\tan 3x = -\sqrt{3}$ is: 1

- (A) $x = \frac{n\pi}{3} - \frac{2\pi}{9}$ where n is any integer
- (B) $x = \frac{n\pi}{3} + \frac{\pi}{3}$ where n is any integer
- (C) $x = \frac{n\pi}{3} - \frac{\pi}{3}$ where n is any integer
- (D) $x = \frac{n\pi}{3} + \frac{2\pi}{9}$ where n is any integer

QUESTION TEN

A ball is thrown into the air from a point O , where $x = 0$, with an initial velocity of 25 m/s at an angle $\theta = \tan^{-1} \frac{3}{4}$ to the horizontal. If air resistance is neglected and the acceleration due to gravity is taken as -10 m/s^2 , then the ball reaches its greatest height after: 1

- (A) 1.5 seconds
- (B) 15 seconds
- (C) $\frac{2}{3}$ of a second
- (D) 3 seconds

_____ End of Section I _____

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet.	Marks
(a) Factorise $a^3 + 27b^3$.	1
(b) Differentiate $\sin^{-1} 5x$.	1
(c) Find $\int \frac{dx}{36 + x^2}$.	1
(d) Evaluate $\int_0^\pi \sin^2 x \, dx$.	2
(e) Find $\int x\sqrt{2 + x^2} \, dx$ using the substitution $u = 2 + x^2$.	2
(f) Solve $\frac{4}{x + 1} < 3$.	2
(g) The variable point $(3t, 4t^2)$ lies on a parabola. Find the Cartesian equation of this parabola.	2
(h) Find the coefficient of x^4 in the expansion of $(3 + x^2)^5$.	2
(i) Prove that $\tan(\frac{\pi}{4} + x) = \frac{\cos x + \sin x}{\cos x - \sin x}$, where $\cos x - \sin x \neq 0$.	2

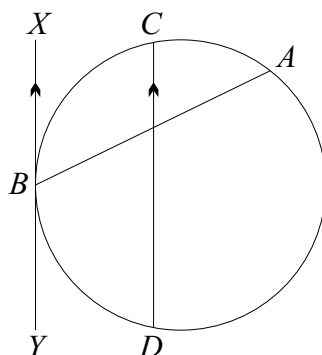
QUESTION TWELVE (15 marks) Use a separate writing booklet.

Marks

(a) Evaluate $\lim_{x \rightarrow 0} \frac{2 \sin 3x}{5x}$.

1

(b)

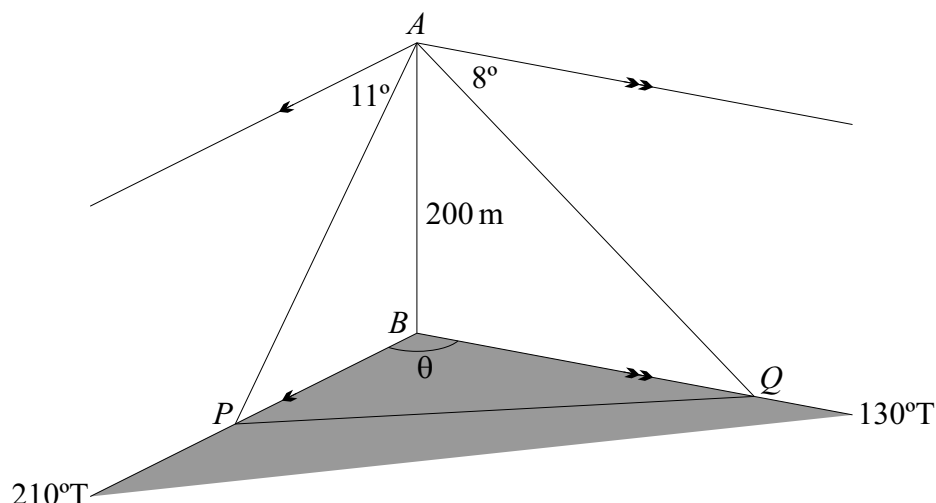


In the diagram above, AB and CD are intersecting chords. The tangent at B is parallel to CD .

3

Copy this diagram into your exam booklet. Let $\angle XBC = \alpha$.
Prove that AB bisects $\angle CAD$.

(c)



HMAS Tarakan is enroute to carry out its latest mission in Cairns. It is first observed from the top of a 200 m cliff, AB at an angle of depression of 8° when it is at the point Q . Ten minutes later it is observed at point P with an angle of depression of 11° . Let $\angle PBQ = \theta$. The bearing of Q from B is 130° T and the bearing of P from B is 210° T.

(i) Show that $PQ^2 = 200^2(\tan^2 79^\circ + \tan^2 82^\circ - 2 \tan 79^\circ \tan 82^\circ \cos \theta)$.

2

(ii) Find the speed of HMAS Tarakan in km/h correct to three significant figures.

3

Exam continues overleaf ...

QUESTION TWELVE (Continued)

(d) (i) Show that $\frac{d}{dx} (e^{4x}(\cos x - 4 \sin x)) = -17e^{4x} \sin x$. 1

(ii) Hence find $\int e^{4x} \sin x \, dx$. 1

(e) The rate at which a body warms in air is proportional to the difference between its temperature T and the constant temperature A of the surrounding air. This rate can be expressed by the differential equation $\frac{dT}{dt} = k(T - A)$ where t is the time in minutes and k is a constant.

(i) Show that $T = A + Be^{kt}$, where B is a constant, is a solution of the differential equation. 1

(ii) An object warms from 5°C to 15°C in 20 minutes. The temperature of the surrounding air is 25°C . Find the temperature of the object after a further 50 minutes have elapsed. Give your answer to the nearest degree. 3

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.

Marks

(a) The polynomial $P(x) = ax^3 - 3x - 1$ has a remainder of -27 when divided by $(x + 2)$.

(i) Show that $a = 4$.

1

(ii) Show that $(x - 1)$ is a factor of $P(x)$.

1

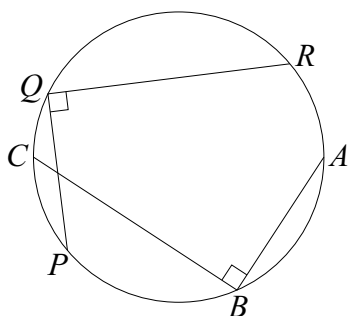
(iii) Hence factorise $P(x)$ fully and sketch the curve $y = P(x)$ showing clearly all intercepts with the axes.

3

(b) Use the principle of mathematical induction to show that $n^3 + 2n$ is divisible by 3 for all positive integers n .

3

(c)



A, B, C, P, Q and R are points on a circle such that $\angle ABC$ and $\angle PQR$ are right angles.

Copy this diagram into your answer booklet.

(i) Explain why $PR = CA$.

1

(ii) Prove that AP is equal and parallel to CR .

2

(d) Consider the identity of $(1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k$.

(i) Show that $1 - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \dots + (-1)^n = 0$.

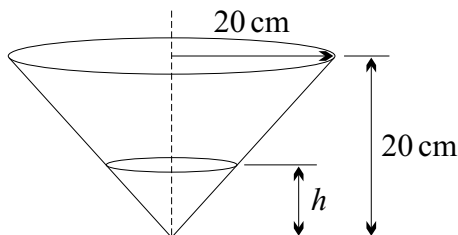
1

(ii) Show that $1 + \frac{1}{2} \binom{n}{1} + \frac{1}{3} \binom{n}{2} + \dots + \frac{1}{n} \binom{n}{n-1} = \frac{2(2^n - 1)}{n + 1}$.

3

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

(a)



A conical vessel has height 20 cm and radius 20 cm. Water is poured into this vessel at a constant rate of 24 cm^3 per second. The depth of water is h cm at time t seconds.

(i) Show that the volume can be written $V = \frac{1}{3}\pi h^3$.

1

(ii) What is the rate of increase of the cross-sectional area A of the surface of the liquid when the depth is 16 cm?

2

(b) A particle moves along the x -axis starting at $x = 0.5$. Its velocity, v metres per second, is described by $v = \sqrt{6x} e^{-x^2}$, where x is the displacement of the particle from the origin.

(i) Find the acceleration of the particle as a function of x .

2

(ii) Find the maximum speed attained by the particle.

2

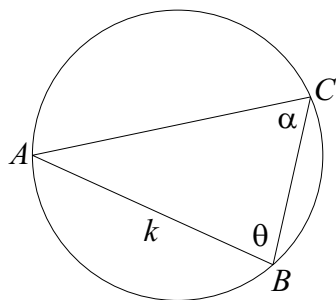
(iii) Show that T , the time taken to travel from $x = 1$ to $x = 3$, can be expressed as

1

$$T = \frac{1}{\sqrt{6}} \int_1^3 x^{-\frac{1}{2}} e^{x^2} dx.$$

Do NOT evaluate this integral.

(c)



Points A , B and C lie on a circle as shown above. The length of the chord AB is a constant k . The sum of the lengths of the chords CA and CB is ℓ . Suppose that $\angle ABC = \theta$ radians and $\angle BCA = \alpha$ radians.

- (i) Show that $\ell = \frac{k}{\sin \alpha}(\sin \theta + \sin(\theta + \alpha))$. **3**
- (ii) Explain why α is a constant. **1**
- (iii) Show that $\frac{d\ell}{d\theta} = 0$ when $\theta = \frac{\pi}{2} - \frac{\alpha}{2}$. **2**
- (iv) Hence show that the maximum value of ℓ occurs when $\theta = \frac{\pi}{2} - \frac{\alpha}{2}$. **1**

_____ End of Section II _____

END OF EXAMINATION

B L A N K P A G E

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SYDNEY GRAMMAR SCHOOL



2012
Trial Examination
FORM VI
MATHEMATICS EXTENSION 1
Thursday 9th August 2012

- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

CANDIDATE NUMBER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

Trial

- | | |
|----|---|
| 1 | D |
| 2 | A |
| 3 | A |
| 4 | B |
| 5 | D |
| 6 | C |
| 7 | C |
| 8 | A |
| 9 | D |
| 10 | A |

Q11.

a) $a^3 + 27b^3 = (a + 3b)(a^2 - 3ab + 9b^2)$. ✓

b) $y = \sin^{-1} 5x$

$$\frac{dy}{dx} = \frac{5}{\sqrt{1-25x^2}} \quad \text{or} \quad \frac{1}{\sqrt{\frac{1}{25}-x^2}} \quad \checkmark$$

c) $\int \frac{dx}{36+x^2} = \frac{1}{6} \tan^{-1} \frac{x}{6} + C \quad \checkmark$

d) $\int_0^{\pi} \sin^2 x \, dx = \frac{1}{2} \int_0^{\pi} (1 - \cos 2x) \, dx \quad \checkmark$
 $= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi}$
 $= \frac{\pi}{2} \quad \checkmark$

e) $\int x \sqrt{2+x^2} \, dx$

$$u = 2+x^2$$
$$du = 2x \, dx$$

$$= \frac{1}{2} \int u^{\frac{1}{2}} \, du \quad \checkmark$$

$$= \frac{1}{2} \cdot \frac{2}{3} u^{\frac{3}{2}} + C$$

$$= \frac{1}{3} (2+x^2)^{\frac{3}{2}} + C \quad \checkmark$$

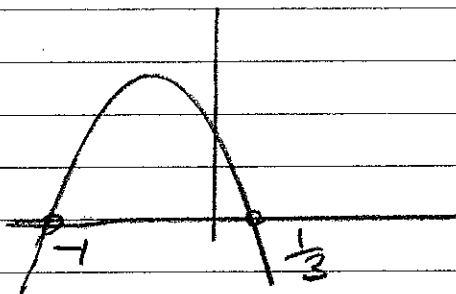
(does not matter if
ext 2 students do not
use sds.)

$$(f) \quad \frac{4}{x+1} < 3$$

$$4(x+1) < 3(x+1)^2$$

$$(x+1)[4 - 3(x+1)] < 0$$

$$(x+1)(1-3x) < 0$$



$$x < -1 \text{ or } x > \frac{1}{3} \quad \checkmark$$

$$(g) \quad x = 3t \quad \text{and} \quad y = 4t^2$$

$$\text{so } t = \frac{x}{3} \quad \text{so} \quad y = 4\left(\frac{x}{3}\right)^2 \quad \checkmark$$

$$y = \frac{4x^2}{9} \quad \text{or} \quad 9y = 4x^2 \quad \checkmark$$

$$(h) \quad {}^5C_2 3^3 (x^2)^2 \text{ is term in } x^4$$

$$\text{So coeff is } {}^5C_2 3^3$$

$$= 270$$

$$(\checkmark \text{ for } {}^5C_2 \text{ and } \checkmark 3^3)$$

$$(i) \quad \text{LHS} = \tan\left(\frac{\pi}{4} + x\right)$$

$$= \frac{\tan \frac{\pi}{4} + \tan x}{1 - \tan \frac{\pi}{4} \tan x}$$

$$= \frac{1 + \tan x}{1 - \tan x} \quad \checkmark$$

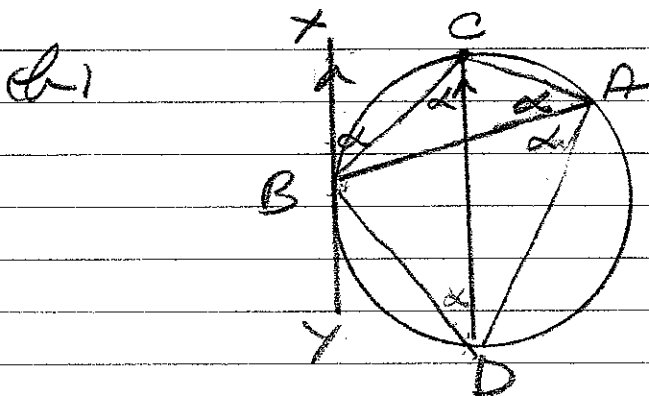
$$= \frac{1 + \frac{\sin x}{\cos x}}{1 - \frac{\sin x}{\cos x}} \times \frac{\cos x}{\cos x} \quad \checkmark$$

$$= \frac{\cos x + \sin x}{\cos x - \sin x}$$

$$= \text{RHS.}$$

Q12.

$$(a) \quad \lim_{x \rightarrow 0} \frac{2 \sin 3x}{5x} = 2 \times \frac{3}{5} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} = \frac{6}{5} \quad \checkmark$$



The angle between tangent and chord equals the angle in the alternate segment. ✓

So $\angle XBA = \angle CAB = \alpha$

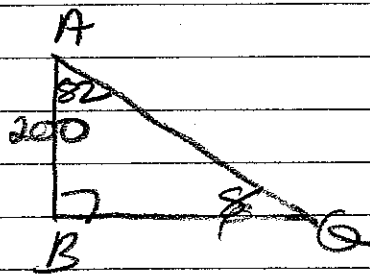
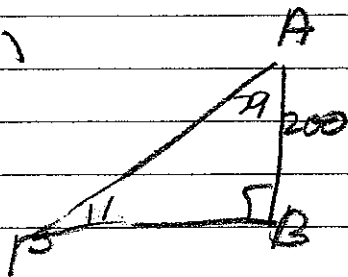
$\angle XBC = \angle BCD = \alpha$, alternate angles, $XY \parallel CD$. ✓

$\angle BCD = \angle BAD = \alpha$, angles in the same segment. ✓

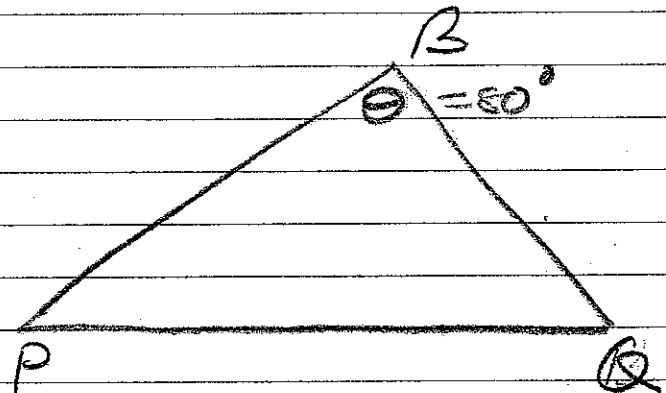
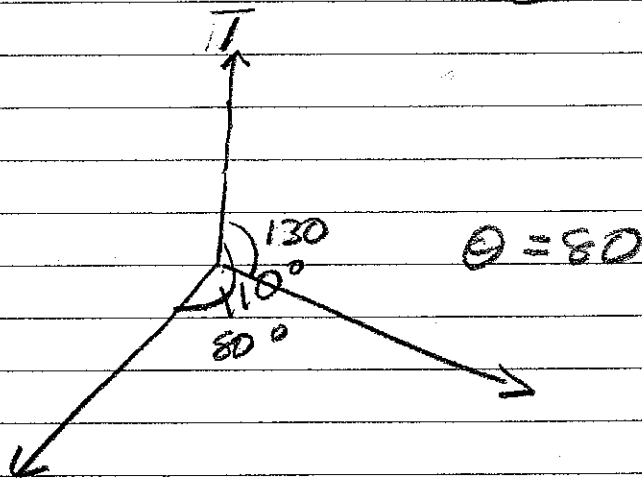
So $\angle BCD = \angle BAD = \alpha$

So AB bisects $\angle CAD$.

(C)



2 marks for showing
where 79° & 82° come
from - either diagram
or writing.



$$(i) \tan 79 = \frac{PB}{200}$$

$$\tan 82 = \frac{BQ}{200}$$

$$PB = 200 \tan 79 \quad \text{and} \quad BQ = 200 \tan 82 \quad \checkmark$$

(ii) Using cosine rule in $\triangle PBQ$.

$$PQ^2 = (200 \tan 79)^2 + (200 \tan 82)^2 -$$

$$2 \times 200 \tan 79 \times 200 \tan 82 \times \cos \theta$$

$$= 200^2 (\tan^2 79 + \tan^2 82 - 2 \tan 79 \tan 82 \cos 80)$$

$$= 2575279.955$$

$$PQ = 1604.76 \text{ m} \quad \checkmark$$

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{PQ}{10} \text{ km per min} \quad \checkmark \text{ for } 80^\circ$$

$$= \frac{1604.76}{10} \times \frac{60}{1000} \text{ km per hr.}$$

$$= 9.6 \text{ km per hr.} \quad \checkmark$$

(e)

(i) $y = e^{4x} (\cos x - 4 \sin x)$

$$u = e^{4x}$$

$$v = \cos x - 4 \sin x$$

$$u' = 4e^{4x}$$

$$v' = -\sin x - 4 \cos x$$

$$\frac{dy}{dx} = vu' + uv'$$

$$= 4e^{4x} (\cos x - 4 \sin x) + e^{4x} (-\sin x - 4 \cos x) \quad \checkmark$$

$$= 4e^{4x} \cos x - 16e^{4x} \sin x - e^{4x} \sin x - 4e^{4x} \cos x$$

$$= -17e^{4x} \sin x$$

(ii) $\int -17e^{4x} \sin x dx = e^{4x} (\cos x - 4 \sin x) + C$

so $\int e^{4x} \sin x dx = -\frac{1}{17} e^{4x} (\cos x - 4 \sin x) + C$

(e)(i) $T = A + Be^{kt}$

$$\frac{dT}{dt} = kBe^{kt}$$

$$= k(T - A)$$

$$Be^{kt} = T - A \quad \checkmark$$

(ii) $t=0, T=5, A=25$

$$5 = 25 + B$$

$$B = -20$$

so $T = 25 - 20e^{kt} \quad \checkmark$

$t=20, T=15$ gives $15 = 25 - 20e^{20k}$

$$-10 = -20e^{20k}$$

$$e^{20k} = \frac{1}{2}$$

$$k = \frac{1}{20} \ln \frac{1}{2} \quad \checkmark$$

$t=70, T = 25 - 20e^{\frac{1}{20} \ln \frac{1}{2} (70)}$

$$\approx 23^\circ$$

Q13.

(a) (i) $P(x) = ax^3 - 3x - 1$

$$\begin{aligned} P(-2) &= -8a + 6 - 1 = -27 \\ -8a &= -32 \\ a &= 4. \end{aligned}$$

(ii) $P(x) = 4x^3 - 3x - 1$

$$P(1) = 4 - 3 - 1 = 0$$

so $(x-1)$ is a factor

(iii) $\begin{array}{r} 4x^2 + 4x + 1. \\ x-1 \end{array}$

$$\begin{array}{r} 4x^3 - 0x^2 - 3x - 1 \\ \underline{4x^3 - 4x^2} \end{array}$$

$$4x^2 - 3x$$

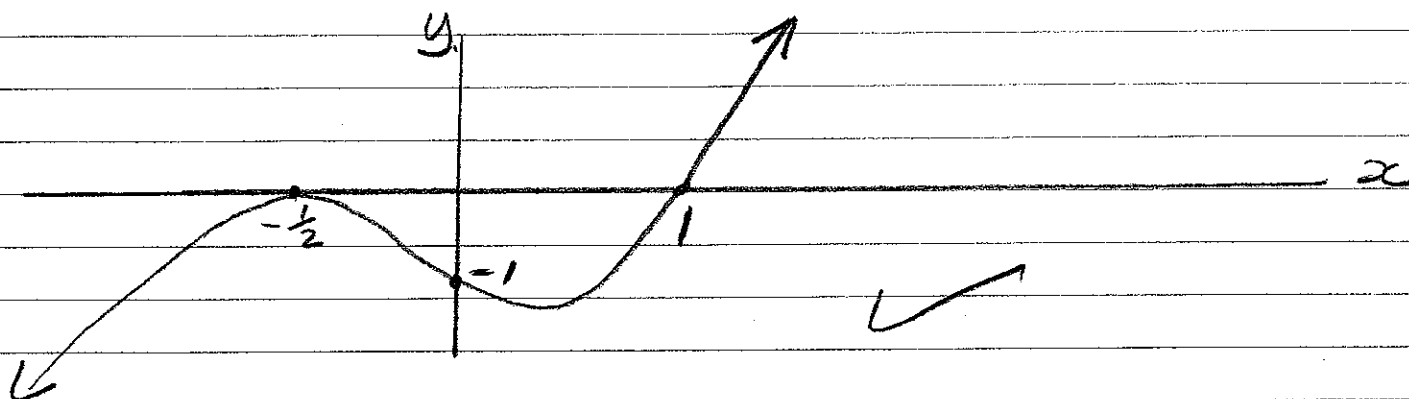
$$\underline{4x^2 - 4x}$$

$$0x - 1$$

$$\underline{x - 1.}$$

now $4x^2 + 4x + 1 = (2x + 1)(2x + 1)$

so $4x^3 - 3x - 1 = (x-1)(2x+1)(2x+1).$



ch
A. Consider $n=1$, $1^3+2=3$ which is divisible by 3
So the statement is true for $n=1$.

B. Suppose that k^3+2k is divisible by 3 for some positive integer k

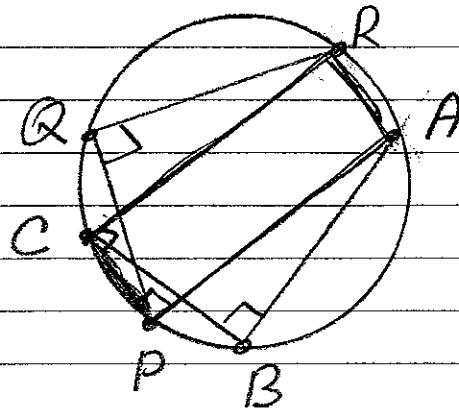
ie suppose that $k^3+2k = 3M$, M an integer

Now prove that $(k+1)^3+2(k+1)$ is divisible by 3.

$$\begin{aligned}\text{Now, } (k+1)^3+2(k+1) &= k^3+3k^2+3k+1+2k+2 \\ &= k^3+2k+3(k^2+k+1) \\ &= 3M+3(k^2+k+1) \text{ using the induction hypothesis} \\ &= 3(M+k^2+k+1) \text{ which is clearly divisible by 3.}\end{aligned}$$

C. So by steps A & B and the principle of mathematical induction, the given statement is true.

C.



(i) PR and CR subtend right angles at the circumference, so they are both diameters, and therefore equal. ✓

(ii) $\angle RCP = 90^\circ$ angle in semicircle, diameter PR.
 $\angle RCQ = 90^\circ$ angle in semicircle, diameter CR.
 $\angle CRA = 180 - 90^\circ$, opposite $\angle CBA$ in cyclic quad CBAR.

So CRAP is a rectangle (3 angles right angles)
So AP is equal and parallel to CR
(opposite sides of rectangle).

✓✓ there are many ways to do this!!!

(d)

$$(i) (1+x)^n = \binom{n}{0}x^0 + \binom{n}{1}x^1 + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n}x^n$$

let $x = -1$
then $0 = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \dots + \binom{n}{n}(-1)^n$

But $\binom{n}{0} = \binom{n}{n} = 1$. ✓

so $0 = 1 - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \dots + (-1)^n$

(ii) integrate both sides of the expansion.

$$\frac{(1+x)^{n+1}}{n+1} = \binom{n}{0}x + \frac{1}{2}\binom{n}{1}x^2 + \frac{1}{3}\binom{n}{2}x^3 + \frac{1}{4}\binom{n}{3}x^4 + \dots + \frac{1}{n+1}\binom{n}{n}x^{n+1} + k. \quad \checkmark$$

Find k : let $x=0$

then $\frac{1}{n+1} = 0 + 0 + 0 + \dots + k.$

so $k = \frac{1}{n+1}$. ✓

and $\frac{(1+x)^{n+1}}{n+1} = \binom{n}{0}x + \frac{1}{2}\binom{n}{1}x^2 + \frac{1}{3}\binom{n}{2}x^3 + \frac{1}{4}\binom{n}{3}x^4 + \dots$

$$\dots + \frac{1}{n}\binom{n}{n-1}x^n + \frac{1}{n+1}\binom{n}{n}x^{n+1} + \frac{1}{n+1}$$

let $x=1$.

$$\frac{2^{n+1}}{n+1} = 1 + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \frac{1}{4}\binom{n}{3} + \dots + \frac{1}{n}\binom{n}{n-1} + \frac{1}{n+1} + \frac{1}{n+1}$$

$$\frac{2^{n+1}}{n+1} - \frac{2}{n+1} = 1 + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \frac{1}{4}\binom{n}{3} + \dots + \frac{1}{n}\binom{n}{n-1} \quad \checkmark$$

$$\frac{2(2^n - 1)}{n+1} = 1 + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \frac{1}{4}\binom{n}{3} + \dots + \frac{1}{n}\binom{n}{n-1}$$

Q14.

(a) (i) Here $r = h$

$$\text{so } V = \frac{1}{3}\pi r^2 h \\ = \frac{1}{3}\pi h^3$$

$$(ii) \frac{dA}{dt} = \frac{dA}{dV} \frac{dV}{dt} \quad \text{and} \quad \frac{dA}{dV} = \frac{dA}{dh} \frac{dh}{dV}$$

$$= \frac{dA}{dh} \frac{dh}{dV} \frac{dV}{dt}$$

$$\text{Now } V = \frac{1}{3}\pi h^3 \quad \text{so} \quad \frac{dV}{dt} = \pi h^2$$

$$A = \pi h^2 \quad \text{so} \quad \frac{dA}{dh} = 2\pi h$$

✓ for
checking the
chain rule
properly.

$$\text{So } \frac{dA}{dt} = 2\pi h \times \frac{1}{\pi h^2} \times 24$$

$$r = 16, \quad \frac{dA}{dt} = \frac{2}{16} \times 24 \\ = 3 \text{ cm}^2 \text{ per sec.}$$

✓

Q-1

(i) $v = \sqrt{6x} e^{-x^2}$

$$v^2 = 6x e^{-2x^2}$$

$$\frac{1}{2} v^2 = 3x e^{-2x^2}$$

✓

$$\begin{aligned} \ddot{x} &= \frac{d(\frac{1}{2}v^2)}{dx} = 3x \cdot (-4x) e^{-2x^2} + 3e^{-2x^2} \\ &= -12x^2 e^{-2x^2} + 3e^{-2x^2} \\ &= 3e^{-2x^2} (1 - 4x^2) \end{aligned}$$

✓

(ii) Fastest speed is when $\ddot{x} = 0$.

$$1 - 4x^2 = 0$$

$$x = \frac{1}{2} \text{ or } -\frac{1}{2} \text{ but } x > 0.5$$

$$\text{so } x = \frac{1}{2}$$

✓

$$x = \frac{1}{2}, \quad v = \sqrt{3} e^{-\frac{1}{4}} \text{ ms}^{-1}$$

✓

(iii) $\frac{dx}{dt} = \sqrt{6x} e^{-x^2}$

$$\frac{dt}{dx} = \frac{1}{\sqrt{6x}} e^{x^2}$$

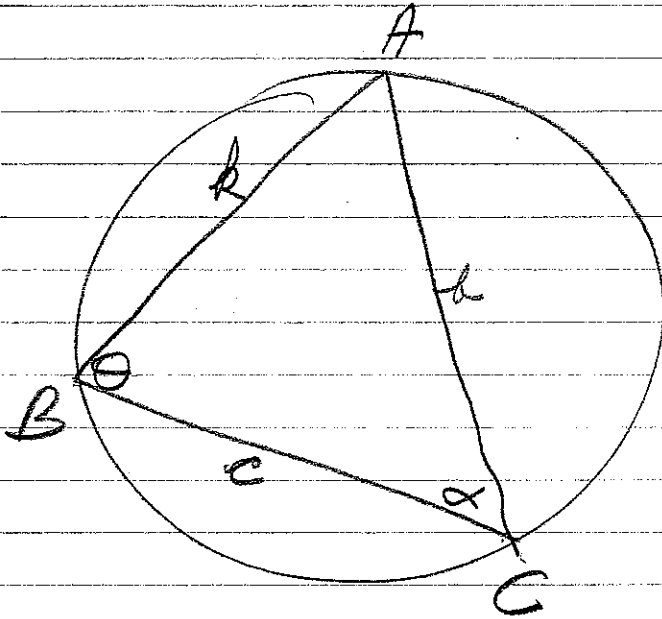
$$T = \int_1^3 \frac{1}{\sqrt{6x}} e^{x^2} dx$$

✓

$$= \frac{1}{\sqrt{6}} \int_1^3 x^{-\frac{1}{2}} e^{x^2} dx$$

14.

(a)



(i) want $(c+b)$

Using sine rule in $\triangle ABC$

$$\frac{b}{\sin \theta} = \frac{k}{\sin \alpha} = \frac{c}{\sin(\pi - \theta - \alpha)}$$

✓ use
of sine
rule.

so $b = \frac{k}{\sin \alpha} \sin \theta$

and $c = \frac{k}{\sin \alpha} \sin(\pi - \theta - \alpha)$

now $b + c =$

$$= \frac{k}{\sin \alpha} (\sin \theta + \sin(\theta + \alpha))$$

✓

$$\left[\begin{aligned} \sin(\pi - \theta - \alpha) &= \sin(\pi - (\theta + \alpha)) \\ &= \sin \pi \cos(\theta + \alpha) - \sin(\theta + \alpha) \cos \pi \\ &= 0 - \sin(\theta + \alpha) \\ &= \sin(\theta + \alpha) \end{aligned} \right]$$

✓

(ii) α is a constant because chord AB is of constant length l , and chords of equal length subtend equal angles at the circumference. ✓

(iii) $l = \frac{R}{\sin \alpha} (\sin \theta + \sin(\theta + \alpha))$.

$$\frac{dl}{d\theta} = \frac{R}{\sin \alpha} (\cos \theta + \cos(\theta + \alpha)) \quad \checkmark$$

at $\theta = \frac{\pi}{2} - \frac{\alpha}{2}$.

$$\frac{dl}{d\theta} = \frac{R}{\sin \alpha} \left[\cos\left(\frac{\pi}{2} - \frac{\alpha}{2}\right) + \cos\left(\frac{\pi}{2} + \frac{\alpha}{2}\right) \right]$$

$$= \frac{R}{\sin \alpha} \left[\cos \frac{\pi}{2} \cos \frac{\alpha}{2} + \sin \frac{\pi}{2} \sin \frac{\alpha}{2} + \cos \frac{\pi}{2} \cos \frac{\alpha}{2} - \sin \frac{\pi}{2} \sin \frac{\alpha}{2} \right] \quad \checkmark$$

$$= \frac{R}{\sin \alpha} [0]$$

$$= 0$$

iv) $\frac{d^2l}{d\theta^2} = \frac{R}{\sin \alpha} (-\sin \theta - \sin(\theta + \alpha)) = -l$

since $l > 0$, this means that $\frac{d^2l}{d\theta^2} < 0$ ✓

so l is a max when $\frac{dl}{d\theta} = 0$ i.e.

$$\theta = \frac{\pi}{2} - \frac{\alpha}{2}$$